

Simon Dovan Nguyen

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EDUCATION

University of Washington, Seattle Ph.D. Candidate in Statistics. Advised by: Tyler H. McCormick	(expected) Jun. 2027
University of Michigan, Ann Arbor M.S. in Applied Statistics. Advised by: Ben B Hansen and Daniel Almirall	Apr. 2023
University of California, Irvine B.S. in Mathematics (Honors); B.A. in Economics (Honors); Minor in Statistics	Jun. 2021

GOOGLE (Sales Applied AI team) | Business Data Scientist Intern Jun. 2026 - Sep. 2026

Deep Reinforcement Learning (RL) for Combinatorial Advertisement Recommendations

- Expanded single-action recommender model capability to multi-action bundling to deliver coordinated advertising strategies.
- Designed RL model to identify complementary ad pairings and optimize bundling synergies, driving high revenue increase.
- Engineered Branching Q-Networks in JAX, cutting complexity from exponential to linear time with <2ms serving latency.

CENTER FOR NAVAL ANALYSIS (Data Science Division) | Data Scientist Intern Jun. 2025 - Sep. 2025

Predictive Aircraft Readiness and Maintenance

- Developed ML models optimizing plane repair & delivery processes, increasing mission-capable F18 Super Hornets to 80%.
- Leveraged NLP to extract latent themes from maintenance logs, enabling earlier detection of supply risks by 12%.

LLM-Powered Tactical Advisor for Specialized Wargaming

- Implemented RAG to dynamically align LLM recommendations with USN wargaming playbooks and established doctrine.
- Outperformed DoD human experts in complex wargaming simulations, demonstrating an 8% reduction in troop casualties.

RESEARCH EXPERIENCE

Reinforcement Learning-Driven Active Learning for Regression [\[Paper\]](#) [\[Codebase\]](#) Jan. 2025 - Present

Under Review at AAAI, 2027

- Improved model performance by up to 8% over state-of-the-art active learning methods across 20 benchmark datasets.
- Utilized Soft Actor-Critic with Potential-Based Reward Shaping to autonomously learn the optimal data sampling policy.

Rashomon-Based Active Learning for Interpretability and Efficiency [\[Proceedings\]](#) [\[Codebase\]](#) Jun. 2024 - Jun. 2026

Published at Conference on Uncertainty in Artificial Intelligence (UAI), 2026

- Improved predictive accuracy by up to 20% with a novel active learning algorithm while reducing labeling cost by over 15%.
- Designed adaptive sampling strategies to strategically collect data that minimize labeling cost while enhancing model training.

Causal Inference & Quasi-Experimental Evaluation in Observational Settings Jun. 2021 - Apr. 2023

- Developed causal inference methods to evaluate intervention effects when randomized A/B testing is infeasible.
- Engineered constrained optimal matching algorithms to balance high-dimensional covariates & reduce selection bias by 9%.

PRESENTATIONS

Conference on Uncertainty in Artificial Intelligence (UAI)	Jul. 2026
Neural Information Processing Systems (NeurIPS) [Abstract]	Dec. 2024
Joint Statistical Meetings (JSM)	Aug. 2023
International Chinese Statistical Association Applied Statistics Symposium (ICSA)	Jun. 2023
Conference on Statistical Practice (CSP) [Abstract]	Feb. 2023
International Conference on Health Policy Statistics (ICHPS) [Abstract]	Jan. 2023

HONORS

Rackham Science Award	Aug. 2023
Rackham Merit Fellowship	Aug. 2021
Departmental Honors in Mathematics and Economics	Jun. 2021

SKILLS

Technical: Python, R, SQL, JAX, PyTorch, pandas, Scikit-Learn, Reinforcement Learning, LLMs, RAG, NLP

Methods: Causal Inference, Quasi-Experimental Design, A/B Testing, Constrained Optimization, Experiment Design, Bandits